任课教师_____ 姓名_____ 学号_____ 班级_____ 学生学院_____

线 订 装

北京交通大学考试试题（A卷）

课程名称：Calculus (B) I 学年学期: 2022—2023 学年第一学期(期末试卷)

课程编号：C108001W 开课学院：数学与统计学院 出题教师：____

题　号	一		二	三	四	五	六	七	八	总分
得　分										
阅卷人										

注意：本试卷共有 6 页，八道大题。

1. (7 point *5)

(1) Find the values of a that make f continuous everywhere.

$$f(x)=\begin{cases}\dfrac{\sin ax}{x} & \text{if } x\neq 0\\ \dfrac{1}{2} & \text{if } x=0\end{cases}$$

(2) Find an equation of the normal line (Cartesian coordinates) of the curve $r=e^{\theta}$ at $\theta=0$.

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(3) Find the length of the curve $x=3t-t^3, y=3t^2, (0\le t\le 1)$.

(4) Evaluate $\int_{-1}^{1}(x^2\sin\frac{\pi}{2}x+\sqrt{1-x^2})dx$.

(5) If xe^x is an antiderivative of $f(x)$, find $\int xf'(x)dx$.

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2. (10point) Determine whether the integral $\int_1^{+\infty} \frac{1}{\sqrt{x}(1+x)} dx$ is convergent or divergent. Evaluate it if it is convergent.

3. (10point) If $f(x)$ is a differentiable function, and

$f(x)\cos x + 2\int_0^x f(t)\sin t\, dt = x+1$. Find $f(x)$.

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4. (10point) Find the area of the region that lies inside both the circle $r=1$ and the cardioid $r=1+\cos\theta$.

5. (10point) Calculate the moments M_x and M_y and the center of mass of a lamina with the density $\rho=1$ and the region bounded by the curves $y=\cos x$, $y=0$, $x=-\frac{\pi}{4}$, and $x=\frac{\pi}{4}$.

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6. (10point) The plane that passes through the point $M(3,1,4)$ and contains the line $x=1+t,\quad y=2t,\quad z=3-t$. Find an equation of the plane.

7. (10point) Through the point $(1,0)$, we do the tangent line L of the curve $y=\sqrt{x-2}$. Let D be the region bounded by the line L, the curve $y=\sqrt{x-2}$ and x-axis. Find the volume of the solid obtained by rotating D about x-axis.

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8. (5point) Let $f(x)$ be the differentiable function on $(-\infty,+\infty)$, and $f'(x)\neq 1$ for $\forall x\in(-\infty,+\infty)$. Show that if $f(x_1)=x_1$ and $f(x_2)=x_2$, $x_1, x_2\in(-\infty,+\infty)$, then $x_2=x_1$.

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